TollRoad — Unit Economics (All-In AWS Cost to Stream)

What does it actually cost to stream one minute of music on TollRoad? This is a bottom-up marginal cost across **every AWS service in the path**, expressed per **minute**, per **song**, and per **user**. All rates are us-east-1 on-demand, **verified 2026-06-12 against the live AWS Pricing API** (`aws pricing get-products` — exact queries in the appendix). Marginal = the incremental cost of one more minute (free tiers are a fixed subsidy, excluded from the per-unit rate; treated separately at the end).

> **Re-modeled 2026-06-18 for the API re-platform.** The serving path is now **Vercel client → API Gateway (REST) → one `tollroad-api` Lambda → Aurora DSQL**, with audio still on **S3 → signed CloudFront** (one mp3 per track, not HLS). The bottom line barely moved — CloudFront egress still dominates — but the middle tier is re-composed: a new API Gateway + Lambda line replaces what used to be Vercel-side metering, the DynamoDB hot path shrank to a best-effort mirror, and Aurora DSQL is now the system of record. The per-minute total went from $0.0001155 to **$0.0001144**; the 10K-MAU bill from $957 to **$948 gross** / **~$846 net**.

The canonical streamed minute

One minute of playback consumes a fixed, known basket of AWS resources:

Dimension	Value	Why
Audio bitrate	**160 kbps → 1.2 MB/min**	Music-quality stream; bitrate is the dominant lever (see below)
CloudFront requests	**~2 / min**	one mp3 per track via signed URL; the browser pulls it in a handful of range GETs (was ~10/min under HLS)
`/v1/charge` calls	**~1 / min**	the metering hot path through API Gateway → Lambda; idempotent per `user#track#minute`
API Gateway requests	**~1.7 / min**	charge + signed-grant refresh + browse/library navigation, averaged over a streamed minute
Lambda invocations	**~1.8 / min**	one `tollroad-api` invoke per API request (256 MB, ~100 ms) + the rare rollup batch
Aurora DSQL	**1 charge txn/min**	`BEGIN → dup-check → debit balance → insert ledger → COMMIT` , the system of record
DynamoDB write units	**~2 / min**	a best-effort METER mirror: 1 conditional `PutItem` + 1 GSI1 (`ALL`) replica, 1/min
DynamoDB stream reads	**$0**	GetRecords by the rollup's Lambda event-source mapping are **not billed**
KMS	**~0 calls/min**	SSE-KMS + CloudFront cache → no per-play decrypt *(requires S3 Bucket Keys — see caveats)*

> **Metering is a dual-write.** `domain/billing.ts` debits the wallet and appends the royalty ledger transactionally in DSQL (durable, authoritative); `domain/meter.ts` then mirrors a single METER event into DynamoDB, whose stream drives the rollup Lambda. Both key on the same `user#track#minute` idempotency key, so the mirror is at-

> least-once-safe and reconciles to a no-op. DSQL is the record of truth; the DynamoDB half is best-effort and skipped entirely in local dev.

Cost per minute streamed — every AWS service

Shown as **$ per 1,000 minutes** (the per-minute figures are fractions of a cent, so 1,000 min is the readable anchor).

AWS service	Unit math	$ / 1,000 min	% of total
CloudFront — egress	1.2 GB × $0.085/GB	**0.10200**	**89.1%**
API Gateway — REST requests	1,714 req × $3.50/M	**0.00600**	**5.24%**
CloudFront — requests	2,000 req × $0.01/10k	**0.00200**	**1.75%**
Aurora DSQL — compute (DPU)	~0.220 DPU/min × $8/M *(modeled)*	0.00176	1.54%
DynamoDB — writes (METER mirror)	2,000 WRU × $0.625/M	0.00125	1.09%
Lambda — API + rollup	~1,811 req + ~45 GB-s	0.00112	0.98%
KMS — key (amortized)	$1/mo ÷ 8.17M min	0.00012	0.10%
S3 — origin GET + storage (amortized)	cache-miss reads only	0.00010	0.09%
DynamoDB — storage (event TTL)	0.3 KB/event, ~30-day resident	0.00006	0.05%
Aurora DSQL — storage (ledger)	100 B/min × $0.33/GB-mo	0.00003	0.03%
DynamoDB — streams	Lambda consumer → GetRecords free	0.00000	0%
TOTAL		**0.11444**	**100%**

All-in marginal cost ≈ $0.0001144 per minute ≈ 0.01144¢ / minute.

Where the money actually goes

- **CloudFront (egress + requests) = ~91%** of the marginal cost. TollRoad is, economically, a **bandwidth business** wearing a database's clothes.
- **The new serving tier — API Gateway + Lambda — is ~6%**, almost all of it API Gateway's per-request fee. It's the largest non-CloudFront line and it's new: under the old architecture this metering ran as Vercel functions, off the AWS bill.
- **The entire data plane — DynamoDB, Aurora DSQL, S3, KMS — is ~3%** combined. Per stream, **the databases are effectively free**; serverless + scale-to-zero means you pay for work done, and the work per minute is tiny.

Rolled up: per song, per user

Unit	Minutes	All-in AWS cost
Per minute	1	**$0.0001144** (0.0114¢)
Per song	3.0	**$0.000343** (0.034¢)
Per user / month	817	**$0.0935** (9.3¢)
Per user / year	9,800	**$1.121**

So a fully-active listener — someone who streams as much as the average Spotify user — costs TollRoad about **$1.12/year, all-in, across the entire AWS stack** — essentially unchanged by the re-platform.

Monthly AWS bill — the stack as deployed

The marginal numbers above, re-expressed as the actual monthly bill the deployed stack (`infra/lib/tollroad-stack.ts`) generates at three scales. Assumes a 10,000-track catalog (~36 GB at 160 kbps) and the always-free tiers (1 TB + 10M req CloudFront, 25 GB DynamoDB storage, 100K DPU + 1 GB-mo DSQL, 1M req + 400K GB-s Lambda). **API Gateway REST has no perpetual free tier** — its 1M-req credit expires after 12 months, so it's a real line from steady state.

Line	Demo (~0 traffic)	1,000 MAU (0.82M min/mo)	10,000 MAU (8.17M min/mo)
CloudFront egress	$0	$0 *(0.98 TB ≤ 1 TB free)*	**$748** *(9.8 TB – 1 free, all in $0.085 tier)*
CloudFront requests	$0	$0 *(1.6M ≤ 10M free)*	$6.34 *(16.3M – 10M free)*
API Gateway (REST)	~$0 *(12-mo free)*	$4.90 *(1.4M req)*	**$49.00** *(14M req, no free tier)*
DynamoDB writes	~$0	$1.02 *(1.6M WRU)*	$10.21 *(16.3M WRU)*
DynamoDB reads + storage	~$0	~$0	~$0 *(reads removed; storage ≤ 25 GB free)*
Aurora DSQL	$0 *(scale-to-zero)*	$0.64 *(180K – 100K free)*	$13.60 *(1.8M DPU – 100K free)*
Lambda	$0	$0 *(free tier)*	$2.76 *(13.8M req; 370K GB-s ≤ 400K free)*
KMS key	$1	$1	$1
KMS decrypt	$0	~$0	$12.26 *(Bucket Keys OFF — see levers; →$0 if ON)*
S3 catalog storage + origin	~$0.85	~$0.90	~$2.50
Total / month	**≈ $2**	**≈ $9**	**≈ $846** *(≈ $0.085/user; ~$834 with Bucket Keys ON)*

The 10K-MAU per-user figure ($0.085) lands just under the marginal 9.3¢ — the gap is exactly the free-tier subsidy. **Until ~1,000 active users the entire AWS bill rounds to the $1/mo KMS key, catalog storage, and a few dollars of API Gateway** (~$9/mo at 1K MAU, vs. ~$4 under the old Vercel-metered topology).

Fixed / amortized AWS lines (not per-minute)

These don't scale with minutes; they amortize to ~$0 per stream but are listed for completeness:

Item	Cost	Per-stream impact
S3 audio storage	a 3-min song @ 160 kbps ≈ 3.6 MB × $0.023/GB-mo = **$0.00008** / **song** / **mo**	divided across every stream of that song → negligible
KMS CMK	**$1** / **mo** per key *version*; the stack enables annual rotation, so **$2** / **mo from year 2** (AWS caps rotated keys at $2)	$0.00000012–24/min amortized (already in the table)
Aurora DSQL / **Lambda idle**	**$0** — both scale to zero	no standing charge
API Gateway idle	**$0** — pay-per-request, no hourly charge	only the per-request fee above
Bedrock "explain my statement"	~1.5k in × $1.10/M + 0.4k out × $5.50/M = **$0.00385** / **call** *(Bedrock on-demand runs ~10% above the direct Anthropic API's $1/$5)*	only when a user asks; ~+4% to a user's monthly cost if used once

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Margin

At an illustrative listener price of **$0.01/min**:

	Per minute	Per user / mo (817 min)
Listener pays	$0.01000	$8.170
All-in AWS cost	$0.0001144	$0.0935
AWS as % of price	**1.14%**	**1.14%**

AWS consumes ~1% of gross billed revenue. But **AWS is not the dominant infrastructure cost — payment processing is.** Both come out of the same platform take, and Stripe-style card fees run **3–7× the AWS bill**.

What the platform actually bears: AWS *and* inbound payments

The platform's cost is **AWS + the inbound processing fee on wallet top-ups**. Standard card processing is **2.9% + $0.30** per charge; the fixed $0.30 makes **top-up frequency** the biggest single lever. Per active listener (gross $98/yr):

Platform cost line	$ / user / yr	% of gross
AWS (all-in, marginal)	~$1.12	~1.1%
Payments — card, monthly top-up	$6.44	6.6%
Payments — card, annual top-up	$3.14	3.2%
Payments — **ACH-first wallet** (0.8%, $5 cap)	**$0.78**	**0.8%**

> **Artist payout is *not* a platform cost.** The artist earns the royalty (90% of gross); the fee to *disburse* it — Stripe Connect / ACH transfer, ~0.25% of payout volume plus any per-account or instant-payout fees — is deducted from the **artist's** share, not the platform rake. So the platform bears AWS + inbound payments only.

At a 10% platform take ($9.80/user/yr of rake), the **kept margin after AWS + inbound payments** swings entirely on the rail and top-up size:

Wallet config	Rake kept
Card, monthly top-up	**23%**
Card, annual top-up	56%
ACH-first, $10 min / auto-reload	**80%**
Pass-through (listener pays the fee)	88%

So the earlier "~88% platform margin" was **AWS-only and optimistic**. All-in, the platform keeps **23–88% of the rake** — and at a 6% rake with monthly card top-ups it goes **negative**. The rake floor is set by **payments, not AWS**. The fix is a **prepaid wallet with a $10 minimum on ACH rails** (bank debit caps the fee), with cards as the instant fallback — see `COST_VISUAL.pdf` pp. 1, 6–7.

Cost levers (in order of impact)

1. **Bitrate** — egress is 89% of cost and scales linearly. 160 → 128 kbps ≈ –18% all-in; 96 kbps ≈ –35%. Quality tiers; default mobile lower.
2. **API Gateway: REST → HTTP API** — the gateway is `apigw.RestApi` at $3.50/M; an HTTP API is **$1.00/M** (`USE1-ApiGatewayHttpRequest`), cutting the new ~$49/mo line to ~$14/mo (–$35/mo at 10K MAU). Highest-ROI change after bitrate, now that API Gateway is the top non-CloudFront line.

3. **S3 Bucket Keys** — collapse per-object KMS requests ~99%, pinning KMS at the flat key fee. **Not yet enabled in the stack** (`bucketKeyEnabled` is unset on the audio bucket) — without it, every cache-miss GET is a billed `kms:Decrypt` (at 10K MAU and a 5% miss rate, ~4M calls ≈ **+$12/mo**). One-line CDK fix.
4. **Edge cache-hit ratio** — long TTLs on immutable audio keep S3 origin GETs and KMS calls near zero; popular catalog is served entirely from the edge.
5. **Charge cadence** — `/v1/charge` runs ~once per streamed minute. Fewer, coarser charges (e.g. settle every 2–3 min) would shave the API Gateway, Lambda, DSQL-txn, and DynamoDB-mirror lines proportionally — at the cost of meter granularity.
6. **Egress volume tiers** — CloudFront drops to $0.08/GB past 10 TB/mo and as low as $0.02/GB at PB scale, so the dominant line *deflates* as the business grows.
7. **Drop the DynamoDB mirror** — DSQL is already the system of record; the METER mirror + Streams + rollup Lambda exist only to feed per-artist/day summaries. Computing those directly in DSQL would retire the ~$10/mo DynamoDB-write line and the whole stream/rollup path. The GSI1 (`ALL`) projection is the bulk of the mirror's WRU; `KEYS_ONLY` would halve it if the mirror stays.

Margin levers (bigger than all of the above):

1. **Payment rail** — ACH/bank debit caps the fee (Stripe 0.8% → $5 ceiling; Adyen $0.26 flat) vs. cards' uncapped 2.9% + $0.30. For a wallet, ACH-first ≈ 4× cheaper than cards → +58 pts of kept rake.
2. **Top-up size** — the fixed $0.30/charge means fewer, larger top-ups win. Monthly → annual card top-ups: $6.44 → $3.14/user/yr. Enforce a **$10 minimum** (keeps even the card fallback under ~6%).
3. **Avoid Stripe metered (per-user monthly invoicing)** — it maxes the fixed-fee count *and* adds the 0.5% Billing fee → 7.1% of gross, the worst option. Keep metering internal (DSQL/DynamoDB); settle aggregates only.

Assumptions & caveats

- **Bitrate 160 kbps** / **1.2 MB per minute** and **3.0 min average song** (consistent with 9,800 min/yr ÷ 3,278 songs ≈ 3 min). Both are direct linear scalars — adjust and every figure moves proportionally.
- **All unit rates verified 2026-06-12 via the AWS Pricing API** (us-east-1 on-demand): CloudFront $0.085/GB + $0.01/10k HTTPS; API Gateway REST $3.50/M (HTTP $1.00/M); DynamoDB $0.625/M WRU, $0.25/GB-mo; DSQL $8/M DPU, $0.33/GB-mo; Lambda $0.20/M + $16.67/M GB-s; KMS $1/key-version-mo + $0.03/10k; S3 $0.023/GB-mo + $0.40/M GET; Bedrock Haiku 4.5 $1.10/M in, $5.50/M out.
- **Serving path**: Vercel is a pure client; every API call proxies to **API Gateway (REST) → one `tollroad-api` Lambda (256 MB) → Aurora DSQL**. Modeled at **~14M API requests/mo** at 10K MAU — ~8.2M `/v1/charge` (1/min) + grant refreshes + browse/library navigation. This load was **off the AWS bill before the re-platform** (it ran as Vercel functions).
- **mp3, not HLS**: audio is one mp3 per track (`audio/<id>.mp3`) behind a signed CloudFront URL. Egress is identical to HLS (1.2 MB/min), but CloudFront *requests* drop from ~10/min (segments) to ~2/min (range GETs). This is the **softest assumption** — a chattier player (more range GETs/seeks) pushes the request line back up; at ~10/min it returns to ~$80/mo and the total lands near $1,013. Egress dominates either way.
- **Metering write path**: `/v1/charge` writes the authoritative balance debit + ledger insert in one DSQL transaction (`domain/billing.ts`), then mirrors one METER event to DynamoDB (`domain/meter.ts`). Hence ~2 WRU per streamed minute in DynamoDB (1 item + 1 GSI1 `ALL` replica), at 1/min cadence — down from ~4 WRU/min under the old DynamoDB-balance model. **No DynamoDB reads remain** in the backend (the dup-check is a DSQL query), so the old live-meter RRU line is gone.
- **DynamoDB Streams cost $0 here**: the only consumer is the rollup Lambda event-source mapping, whose GetRecords calls AWS does not bill. The $0.02/100k rate applies only to non-Lambda consumers.
- **Lambda is now the API tier, not just the rollup**: ~14.8M invocations/mo (one per API request + rare rollup batches), 256 MB, ~100 ms avg → ~370K GB-s, which sits *under* the 400K GB-s perpetual free tier, so net duration is $0 (gross $6.17).
- **Aurora DSQL DPU-per-minute is modeled, not measured** (~0.220 DPU/min, ≈3× the old rollup-only figure to cover the synchronous charge transaction + catalog/library reads). Validate against `AWS/AuroraDSQL` `TotalDPU` CloudWatch metrics under real load. Even at 5× the assumed DPU, DSQL stays under 3% of all-in cost — the conclusion (databases ≈ free per stream) is robust.

- Figures are **marginal** (cost of one more minute). Perpetual free tiers (1 TB CloudFront egress + 10M requests, 25 GB DynamoDB, 100K DSQL DPU + 1 GB-mo, 1M Lambda req + 400K GB-s, 20K KMS requests) make the **first slice of traffic effectively free** — at low volume the real bill rounds to the **$1/mo KMS key** plus a few dollars of un-subsidized API Gateway (see the monthly-bill table).

Appendix — how the rates were verified

All prices pulled live from the AWS Pricing API (us-east-1 endpoint), e.g.:

```
# DynamoDB on-demand / storage / streams
aws pricing get-products --region us-east-1 --service-code AmazonDynamoDB \
  --filters "Type=TERM_MATCH,Field=location,Value=US East (N. Virginia)"

# API Gateway — REST ($3.50/M) vs HTTP ($1.00/M) request pricing
aws pricing get-products --region us-east-1 --service-code AmazonApiGateway \
  --filters "Type=TERM_MATCH,Field=location,Value=US East (N. Virginia)"
# REST  → usagetype USE1-ApiGatewayRequest,     operation ApiGatewayRequest
# HTTP  → usagetype USE1-ApiGatewayHttpRequest

# Aurora DSQL (DPU + storage)
aws pricing get-products --region us-east-1 --service-code AuroraDSQL \
  --filters "Type=TERM_MATCH,Field=location,Value=US East (N. Virginia)"

# CloudFront egress tiers + request pricing
aws pricing get-products --region us-east-1 --service-code AmazonCloudFront \
  --filters "Type=TERM_MATCH,Field=productFamily,Value=Data Transfer" \
            "Type=TERM_MATCH,Field=fromLocation,Value=United States"

# Lambda, S3, KMS — service codes AWSLambda, AmazonS3, awskms (same pattern)

# Bedrock Claude Haiku 4.5 (newer models live under FoundationModels)
aws pricing get-products --region us-east-1 \
  --service-code AmazonBedrockFoundationModels \
  --filters "Type=TERM_MATCH,Field=regionCode,Value=us-east-1"
```

Key dimensions returned (2026-06-12): DDB `WriteRequestUnits $0.625/M`, `Streams $0.0000002/read-request-unit (free for Lambda pollers)`; API Gateway `USE1-ApiGatewayRequest $0.0000035/req (REST)`, `USE1-ApiGatewayHttpRequest $0.000001/req (HTTP)`; DSQL `USE1-DSQL-DistributedProcessingUnits $0.000008/DPU`, `Storage $0.33/GB-Mo`; CloudFront `0-10 TB $0.085/GB`, `HTTPS $0.000001/req`; KMS `$1/key-version-mo`; Bedrock `Claude Haiku 4.5: $1.10/M input, $5.50/M output`.